Ethernet standard and specification

1) Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

Pin	T568A Color	T568B Color
1	White/Green	White/Orange
2	Green	Orange
3	White/Orange	White/Green
4	Blue	Blue
5	White/Blue	White/Blue
6	Orange	Green
7	White/Brown	White/Brown
8	Brown	Brown

RJ-45 Connector (Pins 1 → 8)

Pin: 1 2 3 4 5 6 7 8

T568A W/Green Green W/Orange Blue W/Blue Orange W/Brown Brown

T568B W/Orange Orange W/Green Blue W/Blue Green W/Brown Brown

2) Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Ethernet Standard	**Speed**	**Cable Type**	**Maximum Cable Length**	**Connector Type**
10BASE-T	10 Mbps	Cat3 or higher UTP	100 m	RJ-45
100BASE-TX	100 Mbps	Cat5 or higher UTP	100 m	RJ-45
1000BASE-T	1000 Mbps (1 Gbps)	Cat5e or Cat6 UTP	100 m	RJ-45

3) Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.
   Hint: Think about how transmit and receive pairs are wired in each cable type.

- **Cable to use: Crossover Ethernet cable**

➢ **Reason:**

- A crossover cable swaps the **transmit (TX)** and **receive (RX)** pairs.
- Since both laptops are similar devices (both transmit and receive on the same pins), the crossover cable allows each laptop's transmit signals to connect to the other laptop's receive signals.
- Modern laptops often support **Auto-MDI/MDIX**, which automatically detects the cable type, so a **straight-through cable** may also work. However, if Auto-MDI/MDIX is unavailable, a **crossover cable** is required.

4) List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

Device	**Ethernet Connection**	**Likely Ethernet Standard**
Desktop Computer	Wired LAN port	**1000BASE-T (1 Gbps)**
Wi-Fi Router	LAN/WAN Ethernet ports	**1000BASE-T (1 Gbps)**
Network Printer	Ethernet port	**100BASE-TX** (some newer models support **1000BASE-T**)